SDB-Mamba: Cross-Scale Collaborative Visual State Space Network for Bathymetric Retrieval From Subsurface Reflectance of Satellite Multispectral Imagery

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***Abstract*—Although subsurface-reflectance-based satellite-derived bathymetry (SDB) effectively mitigates sun-glitter effects, current methods struggle to disentangle subtle bathymetric features from complex background environmental noise, severely limiting their generalization across diverse waters. This limitation fundamentally stems from the inability of existing architectures to simultaneously capture high-frequency local textures and long-range semantic dependencies required for precise depth inference in degraded optical environments. To address this challenge, SDB-Mamba, a cross-scale collaborative visual state space (VSS) network, is presented for robust bathymetric retrieval from multispectral imagery. First, a dual-encoder Mamba-ResNet architecture is developed to synergize the linear complexity of VSS models for modeling global contextual dependencies with residual connections for preserving fine-grained local details. Second, to enhance feature discriminability, a spatial attention block (SAB) is integrated to suppress environmental noise in spatially heterogeneous regions, while a channel attention block (CAB) is employed to adaptively recalibrates channel weights based on local optical water characteristics. SDB-Mamba was rigorously evaluated across six diverse study areas with distinct optical properties and bottom types, including unseen domains for external testing. Experimental results demonstrate that the proposed model achieves mean absolute errors (MAEs) of 0.36–0.64 m, reducing root mean square error (RMSE) by up to 86% compared to SwinUNet and improving peak signal-to-noise ratio (PSNR) by 66% relative to TransNet. Ablation studies confirm the critical role of the VSS, SAB, and CAB components, which improved structural similarity index measure (SSIM) by 25%, 68%, and 62%, respectively, establishing SDB-Mamba as a significant advancement in robust, high-precision nearshore bathymetry.**

***Index Terms*—Deep learning, Mamba-ResNet, remote sensing, satellite-derived bathymetry (SDB).**

I. Introduction

SEAWATER is depth data are essential for the applications in such as marine species conservation [1], climate change monitoring [2], marine navigation, and so on [3]. Satellite-derived bathymetry (SDB) retrieval is widely used to obtain seawater depth data [4], [5], [6]. The previous methods have majorly used the surface reflectance (SR) of water (after atmospheric correction) for bathymetry inversion [7]. This type of efforts can be backed to the period from 1970 to 1991 with two major classic categories: 1) semi-empirical model, which refers to incorporate empirical coefficients into optical radiation transfer equation [8], [9], [10], [11], [12], [13]. The major limitation of this type of methods is that they heavily rely either on the spectral relationships of single-pixels or small-zones for bathymetry inversion, making it difficult to perform full-graph context modelingand 2) semianalytical model, which refers to the transformation of the radiation transfer equation [14], [15] into a computable inversion problem by parameterizing the coupling relationship between water depth, intrinsic optical properties (IOPs) [16], and apparent optical properties (AOPs) [17], [18], [19]. Due to the need for pixel-by-pixel computation and complex parameter optimization, this method type is time-consuming and low accuracy of water depth retrieval.

To the problems encountered in traditional methods above, the SDBs based on subsurface of water reflectance have been explored to avoid the interferences of solar glare and the surface roughness, etc. [7], [15]. For example, Chen et al. [20] demonstrated that subsurface reflectivity can be utilized to invert water depth in Class I water bodies. After validation using semianalytical models on GeoEye-1 and Kaneohe Bay imagery, experimental results showed root mean square error

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(RMSE) of 1.18 and 1.34 m, showing slight superiority compared to the traditional Lyzenga empirical model. However, its accuracy is still impacted by water clarity, that is, is not unsuitable for turbid waters. Zhou et al. [7] proposed a novel model for water depth inversion using corrected subsurface reflectance. This approach overcomes the major limitation of existing models. The validation using Landsat 8 imagery under both Class I and Class II waters shows that accuracy of bathymetry inversion increases 13.82% than the early methods.

With the development of artificial intelligence (AI), AI-based bathymetry retrieval models are leveraged to automatically learn the complex relationship between water depth and observed spectral data from remote sensing imagery, enabling high-precision bathymetric inversion [21], [22], [23], [24]. These early methods can be roughly classed into two categories below.

1) *Machine-learning (ML)-based bathymetric inversion,* which can be traced back to Jena et al. [25], who used a radial basis function artificial neural network (RBF-ANN) to predict and plot the depth of seamounts. However, this method is significantly limited in complex seafloor topography. Subsequently, Gholamalifard et al. [26] compared the red/blue single-band algorithm (SBA), principal component analysis (PCA), and multilayer perceptron-ANNs (MLP-ANNs) on Landsat-5 TM data from the southern Caspian Sea. The experimental results showed that MLP-ANNs had the best inversion performance. Kibele and Shears [27] then proposed a purely empirical method based on nonparametric k-nearest neighbors' (KNN) regression to train and validate multibeam sonar data on WorldView-2/3 images, which is better than LLM. Susa [28] compared LLM and LRM with random forest (RF) and extreme gradient boosting (XGBoost) in a horizontal evaluation. They concluded that machine learning methods are overall higher accuracy than traditional linear methods. Saeidi et al. [29] experimented with six machine learning models on the Sentinel-2 dataset and introduced Shapley values (SHAPs) for feature influence analysis. They found that XGBoost performed the best in depth estimation. Although achievements have been reached using ML methods for bathymetry inversion, with limited ability to preserve spatial continuity and geomorphic structure and to generalize across different regions and sensors;
2) *Deep-learning (DL)-based bathymetric inversion,* which has been widely explored in recent years. Li et al. [30] introduced long short-term memory (LSTM) network into bathymetric inversion and demonstrated better performance and mechanistic explanatory power when compared with LLM, LRM, improved LRM, and support vector regression (SVR). Mandlburger et al. [31] constructed a framework that combined laser water depth with multispectral aerial imagery and achieved high-precision inversion through strict geometric correction and U-Net. Knudby and Richardson [32] investigated five types of models (LLM, LRM, RF, dense neural network, and CNN) in a bathymetry performance comparison and pointed out that CNN is optimal in the range of 10–20 m, and multiband and spatial information can significantly improve the accuracy. Agrafiotis and Demir [33] combined the self-attention/cross-attention of U-Net and Swin-Transformer, combined with refraction correction and high-fidelity 3-D reconstruction of the training data, which can not only fill the bathymetric gap, but also serve as an independent SDB scheme, which is significantly better than the U-Net baseline. Liu et. [34] proposed a domain adaptive model (DA-SDB) without atmospheric correction, which achieved the highest accuracy compared with the multiband linear algorithm (MBLA), RF, and a deep neural network in spatiotemporal migration experiments on five sea areas data. The TransBathy based on the Transformer proposed by Zhang and Al Shehhi [35] achieved lower RMSE and higher R^2 in known and unseen areas (including steep slopes, rugged terrain, and other complex terrain), significantly outperforming baseline methods such as LRM, gradient lifting, and CNN, highlighting the advantages of the new generation architecture in complex scenarios. Recent hyperspectral studies have emphasized the importance of spectral enhancement and feature fusion. For example, Li et al. [36] proposed a joint denoising framework to improve spectral representation, while Li et al. [37] introduced a two-stream network for feature fusion. These works indicate that combining spectral refinement with dual-path feature extraction is beneficial for remote sensing tasks. Inspired by these ideas, this study further adapts spectral modulation and dual-stream design for bathymetric inversion. However, the CNN model is constrained by their limited receptive fields, whereas U-Net skips connections, which may propagate shallow noise. When processing multispectral remote sensing imagery, transformers exhibit quadratic growth in computational and memory costs. These growths limit efficient modeling of large-scale subsurface reflectance fields;

As overviewed above, none of the existing SDB methods has proposed a subsurface reflectivity-oriented deep learning model for bathymetry retrieval. For this reason, this article proposes SDB-Mamba, a cross-scale collaborative network model (briefly named "SDB-Mamba model") tailored for shallow-water bathymetry. The main contributions of the proposed SDB-Mamba model include.

1) A cross-scale collaborative dual-encoder framework is proposed that takes satellite multispectral subsurface reflectance images as input and encodes them using Vision Mamba and ResNet-34. Vision Mamba models long-range dependencies and cross-scale context via state-space modeling, while ResNet-34 captures local textures and edge details through hierarchical convolutions.
2) The spatial-channel dual attention mechanism in the adaptive encoder is applied to skip connections to jointly recalibrate spatial and channelwise features of subsurface reflectance image inputs. The spatial branch adaptively highlights fine structures and edges, while

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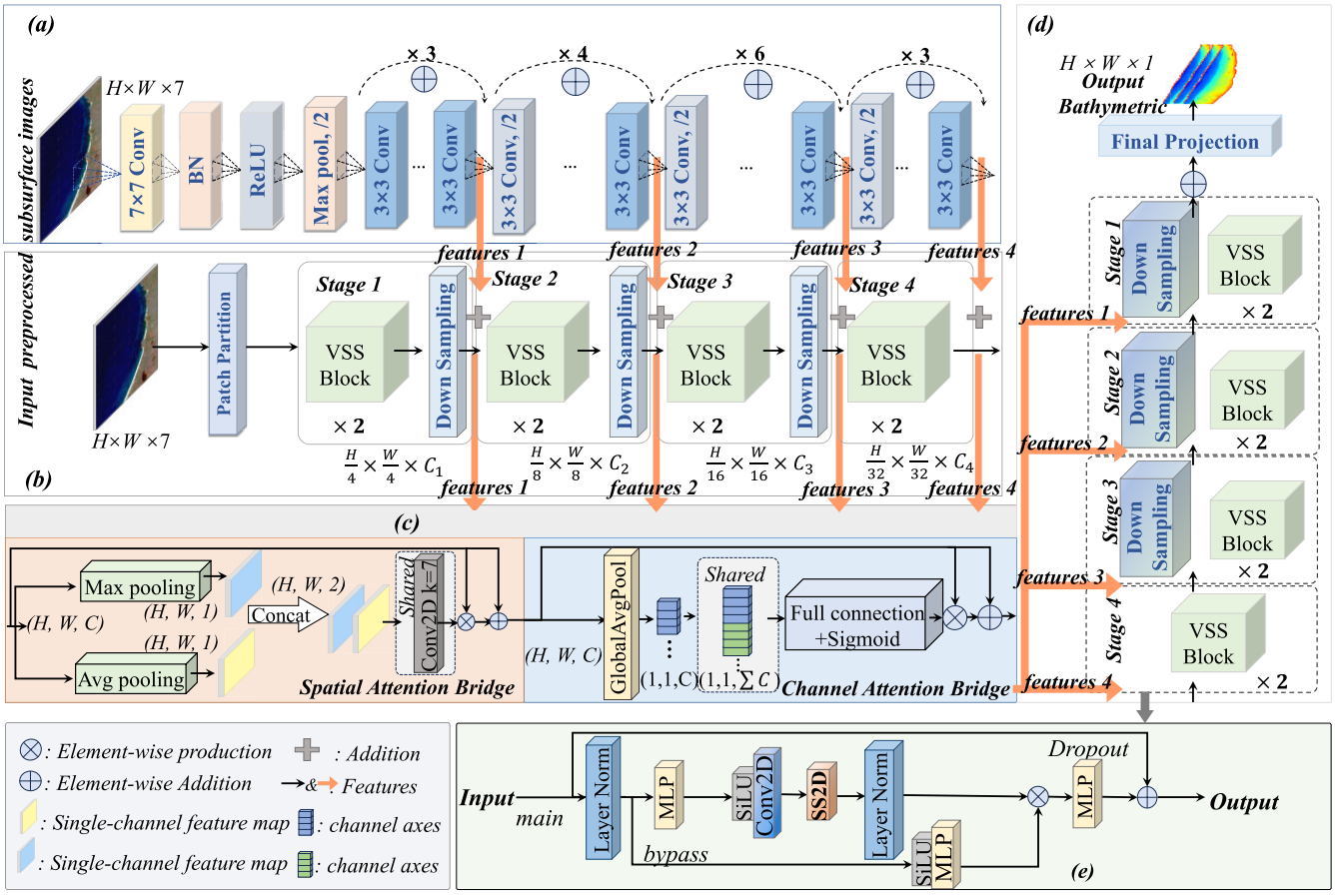


Fig. 1. Architecture of the SDB-Mamba model. (a) Encoder-1 backbone. (b) Encoder-2 backbone. (c) Dual attention bridge. (d) Mamba decoder backbone. (e) Vision state space block.

the channel branch selectively enhances depth-related spectral bands, incorporates subsurface reflectance, and suppresses redundant noise.

II. Proposed Method

To effectively capture the spectral–spatial characteristics of subsurface reflectance for robust bathymetric inversion, this article propose SDB-Mamba, a dual-stream framework built upon the VMU-Net architecture [38], [39]. The main idea of SDB-Mamba is to combine local residual feature extraction with efficient long-range dependency modeling. Specifically, the residual branch preserves fine-scale spatial structures, while the Mamba branch captures global contextual dependencies with linear complexity.

As illustrated in Fig. 1, the proposed framework consists of three main components: 1) a dual-stream encoder that extracts hierarchical multiscale features while preserving correlation-guided spectral priors; 2) a dual-attention feature fusion bridge that selectively recalibrates spatial and channel responses; and 3) a Mamba-based decoder that progressively reconstructs the bathymetric map. Compared with conventional CNN-based or Transformer-based bathymetric inversion networks, the proposed framework explicitly introduces correlation-derived spectral priors and combines them with state-space modeling for efficient local–global feature interaction.

A. Dual-Stream Encoder

The dual-stream encoder is designed to jointly capture local spatial details and global contextual dependencies. Encoder-1 focuses on correlation-guided multiscale residual feature extraction, while Encoder-2 models long-range spatial dependencies using the Vision State Space architecture. Both encoders take the same modulated input to ensure spectral consistency across the two feature paths.

1) Correlation-Aware Spectral Modulation: The purpose of this module is to emphasize spectral bands that are more informative for shallow-water bathymetry (Fig. 2). Unlike conventional channel attention modules that learn spectral weights implicitly from feature maps, the proposed modulation directly constructs a normalized spectral weight vector from wavelength-wise Spearman–Wilson correlations. This design introduces domain-specific spectral priors before deep feature extraction. Given the subsurface reflectance input X_{sub}, the modulated input is obtained by

$$X_{mod} = X_{sub} \odot \boldsymbol{w} \tag{1}$$

where $w \in \mathbb{R}^C$ denotes the normalized correlation-derived spectral weight vector, and $\odot$ represents channelwise multiplication. The Coastal–Blue–Green bands usually receive larger weights because they provide stronger water-penetrating signals for shallow-water bathymetric retrieval. The modulated

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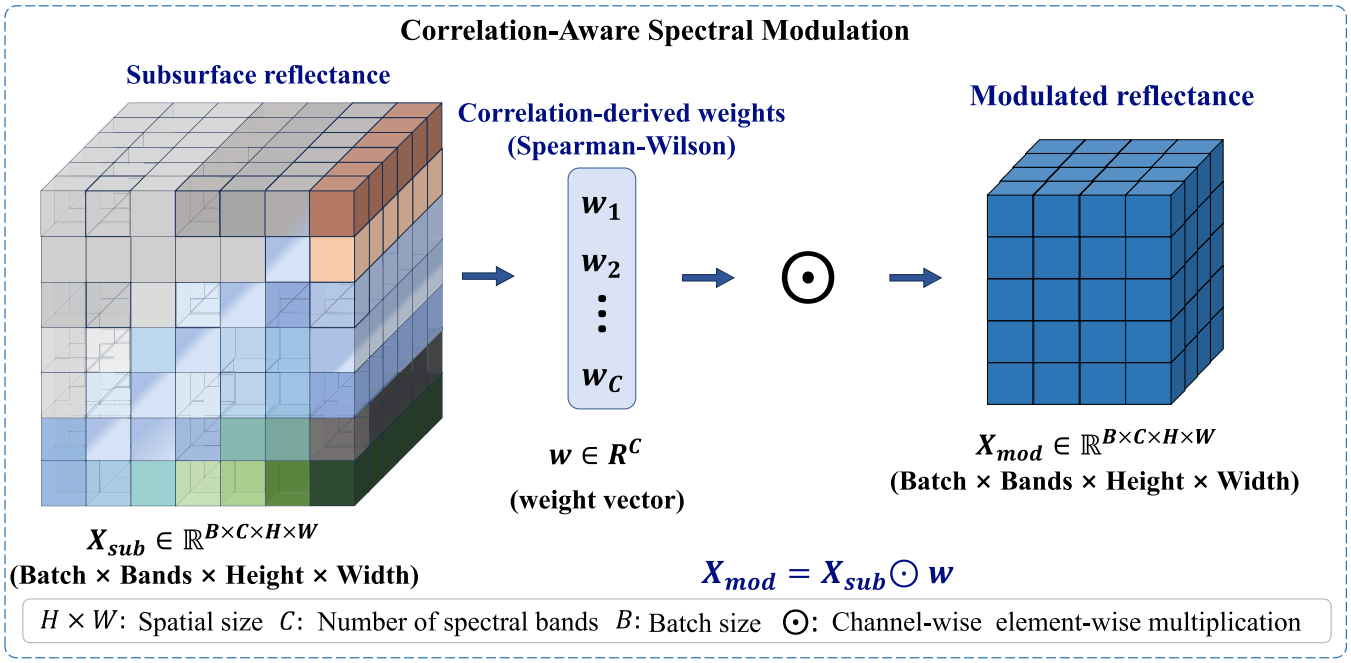


Fig. 2. Correlation-aware spectral modulation.

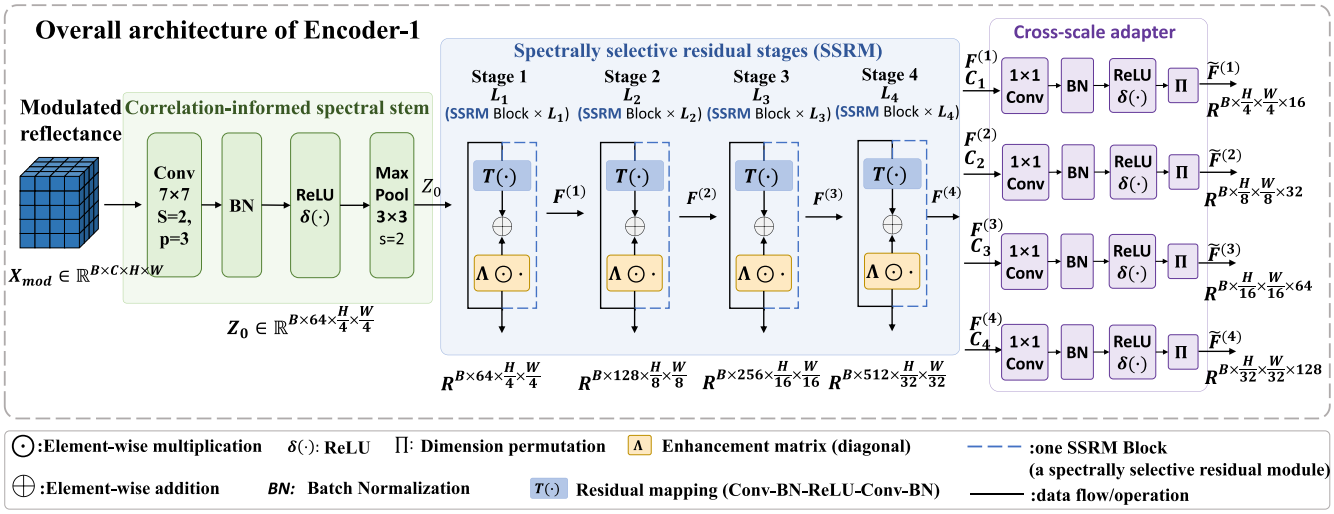


Fig. 3. Overall architecture of Encoder-1.

input X_{mod} is then fed into both Encoder-1 and Encoder-2 as a unified input.

2) Encoder-1: Correlation-Guided Multiscale Residual Extraction: Encoder-1 [Fig. 1(a)] aims to extract hierarchical spatial features while preserving the correlation-guided spectral prior introduced in (1). Different from a standard ResNet backbone, Encoder-1 introduces a spectrally selective residual mechanism (SSRM) into the shortcut path, so that depth-relevant spectral responses can be maintained during deep feature propagation.

a) Correlation-informed spectral stem: The modulated input X_{mod} is first processed by a stem module to generate the initial feature embedding (Fig. 3). This stem reduces the spatial resolution and extracts low-level spatial–spectral features

$$Z_0 = MaxPool\left(\delta\left(BN\left(Conv_{7\times 7,s=2p=3bias=0}\left(X_{mod}\right)\right)\right)\right) \quad (2)$$

where δ denotes the ReLU activation function. The intermediate feature embedding Z_0 is explicitly labeled in Fig. 3 and serves as the input to the first residual stage.

b) Ssrm: After the stem, Z_0 is passed through four residual stages $\{L_i\}_{i=1}^4$ to obtain hierarchical representations. To prevent the correlation-guided spectral prior from being weakened in deep layers, this article introduces a learnable enhancement matrix Λ into the residual shortcut. Unlike standard residual connections that treat all channels equally, the proposed shortcut adaptively reweights spectral feature channels.

The refined residual mapping is defined as

$$F^{(\iota)} = \delta\left(\left(\mathcal{T}\left(F^{(\iota-1)}\right) + \left(\Lambda \odot F^{(\iota-1)}\right)\right)\right), \quad F^{(0)} = Z_0 \quad (3)$$

where $\mathcal{T}(\cdot)$ denotes the residual convolutional mapping, and $\odot$ denotes the Hadamard product with channelwise broadcasting.

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The enhancement matrix is formulated as

$$\Lambda = diag(), \quad = [_1, _2 \ldots _C]^T$$

where contains learnable channelwise enhancement parameters. To incorporate the spectral prior from (1), is initialized by the normalized correlation weights w. This initialization encourages the network to preferentially propagate features associated with informative water-penetrating bands while allowing the weights to be further optimized during training.

Algorithm 1 Correlation-Guided Hierarchical Dual-Encoder Fusion Strategy

Input: Subsurface reflectance X_{mod} (7 channels)
Output: Fused multi-scale features $\{E^{(1)}, E^{(2)}, E^{(3)}, E^{(4)}\}$

Step 1: Correlation-aware spectral modulation
Calculate weight $\boldsymbol{w}$ (Spearmen-Wilson)
$X_{mod} \leftarrow X_{sub} \odot \boldsymbol{w}$ (Comment: Prioritize Coastal/Blue/Green)
Step 2: Dual-path initialization
$Z_0 \leftarrow Stem(X_{mod})$
$F^{(0)} \leftarrow Stem(X_{mod})$ (Encoder-1)
$E^{(0)} \leftarrow PatchEmbed(X_{mod})$ (Encoder-2)
Step 3: Spectrally selective residual learning
Initialize $\Lambda = diag()$ **, where** $\lambda_{\text{init}} = \mathbf{w}$
For *stage*$\iota = 1$ to 4 **do**:
$F^{(\iota)} \leftarrow ResBlock_\iota(F^{(\iota-1)}) + (\Lambda \odot F^{(\iota-1)})$
$\tilde{F}^{(\iota)} \leftarrow \text{Adapter}(F^{(\iota)})$
$E^{(\iota)}_{global} \leftarrow VSSBlock_\iota(E^{(\iota-1)})$
$E^{(\iota)} \leftarrow \tilde{F}^{(\iota)} + E^{(\iota)}_{global}$
End For
Return $\{E^{(1)}, E^{(2)}, E^{(3)}, E^{(4)}\}$
End Procedure

c) Cross-scale adapter: The output of each residual stage is compressed and reshaped before being injected into the Mamba path. This adapter differs from direct feature addition because it aligns both the channel dimension and memory layout between the residual branch and the sequence branch. Specifically, a 1×1 convolution is used for channel compression, followed by batch normalization, ReLU activation, and a permutation operator Π:

$$\tilde{F}^{(\iota)} = \Pi\left(\delta\left(BN\left(W^{(\iota)}_{1\times1} * F^{(\iota)}\right)\right)\right) \tag{4}$$

where $W^{(\iota)}_{1\times1}$ denotes the learnable 1×1 convolution kernel. The overall correlation-driven hierarchical dual-encoder fusion procedure is summarized in Algorithm 1. The compressed features have channel dimensions {16, 32, 64, 128} and are used for subsequent interaction with Encoder-2.

3) Encoder 2:

a) Global dependency modeling with vision mamba: Encoder-2 is designed to capture long-range spatial dependencies that are difficult to model using local convolution alone. As shown in Fig. 1(b), this branch adopts the visual state space (VSS) architecture to model global contextual information. Compared with transformer-based encoders that rely on self-attention with quadratic computational complexity, the Vision Mamba branch achieves efficient global context modeling through state-space modeling with linear complexity.

b) Patch embedding: The modulated input X_{mod} is transformed into a visual token sequence through a patch embedding layer, as illustrated in Fig. 1(b). Unlike standard patch embeddings that directly project the original input, this step projects the re-weighted spectral signals into a latent dimension C, where the effective water-information window, including the Coastal–Blue–Green bands, has been emphasized

$$E^{(0)} = PatchEmbed_{4\times4}(X_{mod}), \quad E^{(0)} \in \mathbb{R}^{\frac{H}{4}\times\frac{W}{4}\times C} \tag{5}$$

where the patch size is set to 4×4. This tokenization allows the model to process bathymetric information as a sequence while preserving the correlation-guided spectral priorities established in the input stage.

c) Vss block and ss2d mechanism: The core engine of the VSS block is the 2-D-Selective Scan (SS2D) module, which bridges 1-D sequential modeling with 2-D spatial structures [Fig. 1(e)]. As illustrated in Fig. 4, the input tokens are unfolded along four scanning directions to enable multidirectional spatial interaction. Specifically, the scan-expanding process is shown in Fig. 4(a). Specifically, the scan-expanding process is shown in Fig. 3(a), the S6 state-space update is shown in Fig. 4(b), and the scan-merging process is shown in Fig. 4(c).

The sequential modeling process is based on a discretized state space model. The continuous system is discretized using the zero-order hold (ZOH) rule to process the bathymetric sequence x_t derived from the unfolded patches

$$h_t = \bar{\boldsymbol{A}}h_{t-1} + \bar{\boldsymbol{B}}x_t, y_t = Ch_t \tag{6}$$

where $x_t \in \mathbb{R}^D$ represents the bathymetric feature vector at step t, and h_t is the hidden state that dynamically filters environmental noise, and $\bar{\boldsymbol{A}}$, $\bar{\boldsymbol{B}}$ are discretized parameters [41], [42]. After selective scanning, the processed sequences are reconstructed into the original 2-D layout through scan merging, as shown in Fig. 4(c). By adaptively updating h_t, the model captures long-range contextual patterns, such as coastal gradients and smooth depth transitions, with linear computational complexity.

d) Hierarchical dual-path fusion: As the sequence propagates through the network stages, the global contexts captured by the VSS blocks are progressively enriched by local spatial details from Encoder-1. The detailed fusion procedure is summarized in Algorithm 1. At the ι-th stage, the compressed spatial feature $\tilde{F}^{(\iota)}$ from [see (4)] is injected into the Mamba path

$$E^{(\iota)} = VSSBlock_\iota\left(E^{(\iota-1)}\right) + \tilde{F}^{(\iota)} \tag{7}$$

where $E^{(\iota)}$denotes the fused feature sequence at stage ι and serves as the input to the subsequent stage. Here, $VSSBlock_\iota$ models the global dependency from the previous state, while the elementwise addition of $\tilde{F}^{(\iota)}$ introduces local spatial details from Encoder-1. This fusion strategy improves local–global consistency in bathymetric prediction.

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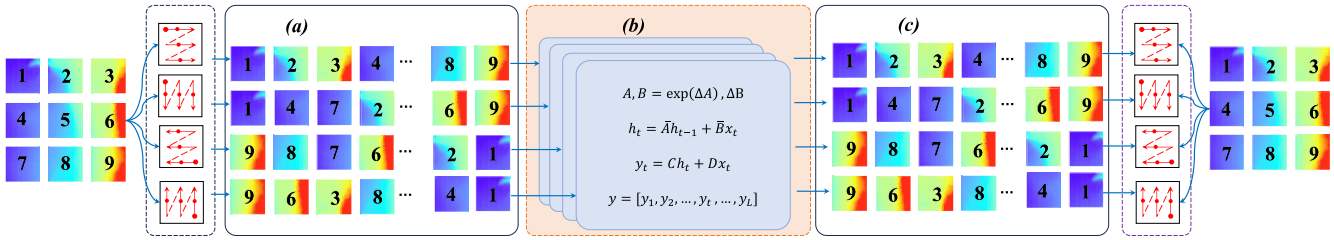


Fig. 4. SS2D architecture chart. (a) Scan expanding. (b) S6 block. (c) Scan merging.

Algorithm 2 Dual-Attention Feature Fusion Process

Input: Feature map $\boldsymbol{F}_{in}$ 2
Output: Refined feature map $\boldsymbol{F}_{out}$
Parameters: Kernel size $k = 7$, MLP weights

Procedure Dual-Attention ($\boldsymbol{F}_{in}$)
Step 1: Spatial Attention Module (SAB)
$\boldsymbol{F}^S_{avg} \leftarrow AvgPool(\boldsymbol{F}_{in})$
$\boldsymbol{F}^S_{max} \leftarrow MaxPool(\boldsymbol{F}_{in})$
$\boldsymbol{F}_{concat} \leftarrow Concat([\boldsymbol{F}^S_{avg}; \boldsymbol{F}^S_{max}])$
$\boldsymbol{M}_s \leftarrow \sigma(Conv^{7\times7}(\boldsymbol{F}_{concat}))$
$\boldsymbol{F}_{sp} \leftarrow \boldsymbol{M}_s \otimes \boldsymbol{F}_{in} + \boldsymbol{F}_{in}$
Step 2: Channel Attention Module (CAB)
$\boldsymbol{v} \leftarrow GlobalAvgPool(\boldsymbol{F}_{sp})$
$\boldsymbol{F}_{out} \leftarrow \boldsymbol{M}_c \otimes \boldsymbol{F}_{sp} + \boldsymbol{F}_{sp}$
return $\boldsymbol{F}_{out}$
End Procedure

B. Dual-Attention Feature Fusion Bridge

To mitigate feature redundancy and enhance informative spatial–spectral responses in multispectral remote sensing imagery, a dual-attention mechanism is embedded within the skip connections for adaptive feature refinement. As shown in Fig. 1(c), this bridge acts as a feature refiner that recalibrates encoder features before they are integrated into the decoder stages. Unlike ordinary skip connections that directly transfer encoder features, the proposed bridge selectively emphasizes bathymetry-relevant spatial regions and spectral channels.

1) *Spatial Attention Block (SAB):* Following Algorithm 2, the SAB captures spatially informative regions from the input feature map. As depicted on the left side of Fig. 1(c), parallel global max pooling and average pooling are applied along the channel dimension [43]. The pooled descriptors are concatenated and processed by a 7×7 convolution to generate a spatial attention map. This map is multiplied with the original feature map to highlight bathymetry-relevant regions, such as shoreline boundaries, depth transition zones, and sediment changes.
2) *Channel Attention Block (CAB):* After spatial refinement, the CAB models interchannel correlations. It utilizes global average pooling to compress spatial dimensions into a channelwise descriptor [43]. The descriptor is then passed through a shared mapping network to generate channel weights, which recalibrate the feature responses. This process enhances spectral channels that are more relevant to water depth estimation.
3) *Integration and Shortcut:* Both attention blocks incorporate residual connections to maintain training stability, as summarized in Algorithm 2. The refined features are then delivered to the corresponding hierarchical levels of the Mamba-based decoder, as indicated by the multiscale skip connections in Fig. 1. This design reduces redundant responses in subsurface multispectral features and provides a more robust feature hierarchy for spatially coherent bathymetric reconstruction.

C. Mamba-Based Decoder

The Mamba-based decoder progressively restores spatial resolution by integrating high-level semantic features with the refined skip features from the dual-attention bridge. As shown in Fig. 1(d), the decoder follows a hierarchical reconstruction process with patch expansion, feature fusion, VSS-based refinement, and final projection. Compared with conventional convolutional decoders, this decoder uses VSS modules to preserve long-range spatial dependencies during resolution recovery.

1) *Spatial Reconstruction With VSS Modules:* At each decoding stage, the upsampled deep features are fused with the refined skip features generated by the dual-attention bridge in Algorithm 2. The fused features are then processed by VSS modules, where the SS2D mechanism models long-range dependencies [44]. This process helps recover spatially coherent seabed structures and reduces information loss caused by downsampling.
2) *Resolution Recovery and Projection:* Patch expansion layers are used as the upsampling units, which progressively increase the spatial resolution and reduce the channel dimension at each stage. After the feature map is restored to the original spatial size $H \times W$, a 1×1 projection convolution maps the high-dimensional representation to a single-channel depth map. The final depth prediction $\hat{Y}$ is formulated as

$$\hat{Y} = \sigma\left(F_{proj}\left(\boldsymbol{D}_{final}\right)\right) \tag{8}$$

where F_{proj} represents the projection layer, and σ denotes the Sigmoid activation function to constrain the depth values. The entire end-to-end workflow-comprising dual-path encoding, attention-based refinement, and Mamba-based decoding, is formalized in Algorithm 3.

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TABLE I
CHARACTERISTICS OF THE STUDY AREAS

#	Locations	Geographic coordinates	Climates	Seafloor Types	Satellite image available
Study area-1	Southeast side of Oahu Island, Hawaii, USA	21°22'N, 157°40'W	Warm and humid subtropical climate	Coral reefs, sandy beaches, lagoons, seagrass beds, volcanic rock headlands	2013.10-2025.03
Study area-2	West side of Molokai Island, Hawaii, USA	21°08'N, 157°02'W	Tropical climate	Crystal-clear tropical waters, gently sloping terrain	2013.05-2025.06
Study area-3	West of Weizhou Island, Guangxi, China	21°02'N, 109°06'E	South subtropical monsoon climate	Gentle slope seafloor topography	2013.11-2025.05
Study area-4	West of Xuwen County, Guangdong Province, China (Coral Reef Conservation Area)	20°22'N, 109°51'E	Tropical monsoon climate	Coral reef ecological integrity	2021.02-2025.06
Study area-5	West of Tung Ping Chau Island, Hong Kong, China	22°33'N, 114°26'E	Subtropical maritime climate	Shale-dominated with well-defined horizontal bedding	2021.09-2025.01
Study area-6	Northwest of Saipan Island, Northern Mariana Islands, United States	15°12'N, 145°51'E	Tropical maritime climate	White sand, coral reefs, reef flat rocky substrate.	2014.10-2024.05

Algorithm 3 SDB-Mamba End-to-End Bathymetry Estimation

Input: 7 channels subsurface spectrum image $\mathbf{I} \in \mathbb{R}^{7\times H\times W}$
Output: Predicted depth map $\hat{Y}$

Procedure SDB-Mamba ($\mathbf{I}$)
Phase 1: Dual-encoder fusion
  $\{E_1, E_2, E_3, E_4 \leftarrow \text{DualEncoder}(\mathbf{I})\}$

Phase 2: Attention-based refinement
  for $i = 1$ **to** 3 **do**
    $\mathbf{E}_i^{'} \leftarrow DualAttention(\mathbf{E}_i)$
  end for

Phase 3: Mamba-based decoding
  $\mathbf{D_4} \leftarrow \mathbf{E_4}$
  for $i = 3$ **down to** 1 **do**
    $\mathbf{D}_{up} \leftarrow PatchExpand(\mathbf{D}_{i+1})$
    $\mathbf{D}_{fused} \leftarrow Concat(\mathbf{D}_{up}, \mathbf{E}_i^{'})$
    $\mathbf{D}_i \leftarrow VSS\,Block(\mathbf{D}_{fused})$
  end for

Phase 4: Depth Regression
  $\mathbf{D}_{final} \leftarrow FinalExpand(\mathbf{D}_1)$
  $\hat{Y} \leftarrow \sigma(Conv^{1\times1}(\mathbf{D}_{final}))$
return $\hat{Y}$
End Procedure

III. EXPERIMENTS AND ANALYSIS

A. Six Study Areas and Their Different Characteristics

This study selects six study areas representing diverse coastal geomorphologies and optical water conditions to evaluate the performance and transferability of the proposed method. The six study areas span the North Pacific, the West Pacific, the Beibu Gulf in Guangxi, China, the South China Sea, and Dapeng Bay in Shenzhen, China (see Figs. 5 and 6). Table I summarizes their characteristics.

B. Data Preprocessing

1) Spectrum Data Preprocessing:

To reduce the influence of atmospheric correction residuals and radiometric calibration uncertainties, 120 Landsat-8 SR [45] scenes covering the study area from 2013 to 2025 with overall cloud cover < 10% were selected (Table I), and the quality assessment band was used to mask cloud, cloud-shadow, and land pixels (Fig. 3).

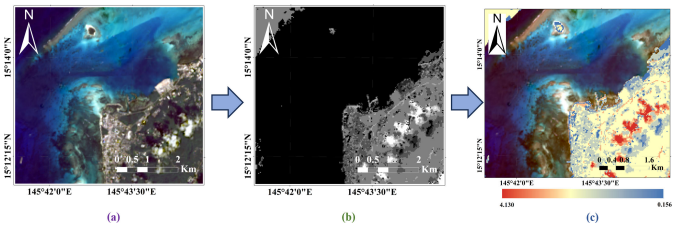


Fig. 5. Flowchart for spectral data preprocessing. (a) Original satellite image. (b) Marine water body extraction. (c) Subsurface water body correction.

a) Water Body Mask Construction: This study employs L2-level imagery to construct the integrated normalized difference water index (INDWI), shown in Fig. 5(b) as defined by the following:

$$\text{INDWI} = \frac{1}{2}\left(\frac{B_G - B_N}{B_G + B_N} + \frac{B_G - B_{S2}}{B_G + B_{S2}}\right) \tag{9}$$

where B_G represents the green band, B_N represents the near-infrared band, and B_{S2} represents the shortwave infrared 2 band. The index ranges from –1 to 1.

b) Correction of multispectral subsurface reflectance in space: To mitigate sun glint contamination, where specular reflection compromises the spectral integrity of the water body, shown in Fig. 5(c), this study implements a subsurface reflectance correction algorithm [7]

$$r_{rs}(i) \approx \frac{2\left(R_{rs}^{S}(i) + R_{rs}^{n}(5)\right)}{1 + 3\left(\left(R_{rs}^{S}(i) + R_{rs}^{n}(5)\right)\right)} \tag{10}$$

where $R_{rs}^{S}(i)$ represents the smoothed SR image for band i, obtained by subtracting the original reflectance of band i from band 5; $R_{rs}^{n}(5)$ represents the "true" value of the "noise-free" SR in band 5, calculated by multiplying the smoothed SR in band 4 by an empirical proportional coefficient and adding an empirically fit offset value, where $i \in \{1, 2, 3, 4, 6, 7\}$.

c) Depth data preprocessing: High-quality in situ bathymetric data is used to serve as the ground truth for model training, validation, and performance assessment.

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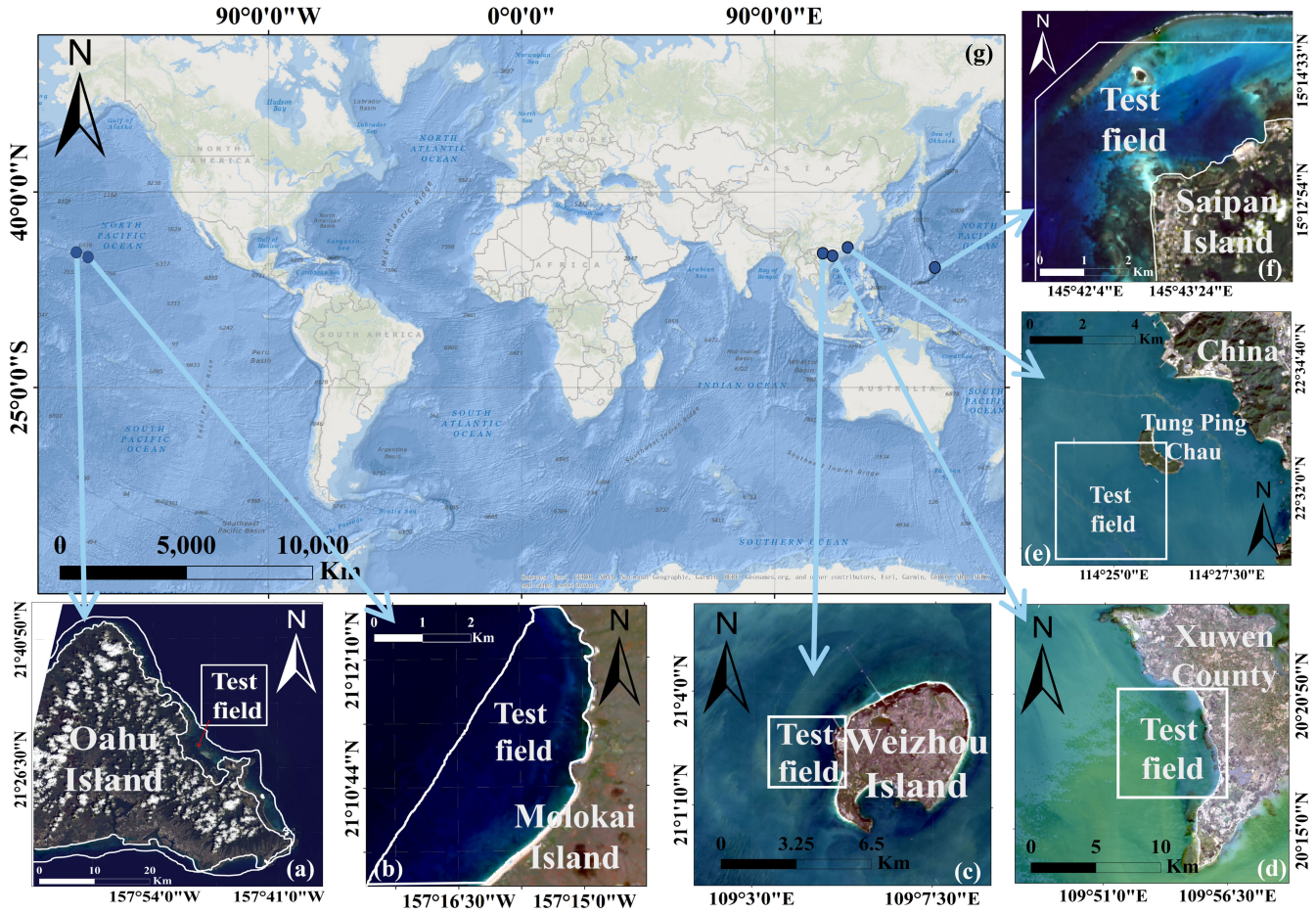


Fig. 6. Six study areas, which are (a) Oahu Island, (b) Molokai Island, (c) Weizhou Island, (d) Tung Ping Chau, (e) coastal area of Xuwen County, (f) Saipan Island, and (g) area cropped from NOAA's global ocean map.

TABLE II
MEASUREMENT ACCURACY OF THE RESON SEABAT 7125 MULTIBEAM BATHYMETRIC SYSTEM

Plane Accuracy (95% Confidence)	Sounding Accuracy (95% Confidence)	100% Seafloor Scanning	System Detection Capability
2m	0.25m	It has to be done	Characteristics of space Objects>1m^3

TABLE III
ERROR LIMITS OF THE POSITION ERROR OF SOUNDING POSITIONING POINT

0-20m	**20-30m**
±0.3m	±0.4m

TABLE IV
MAIN PARAMETERS OF THE CZMIL HYDROLOGICAL MODEL

Pulse repetition frequency	10kHz
Maximum depth single pulse	Kd·Dmax=3.75-4.0 daytime (bottom reflectivity>15%)
Minimum depth	<0.15m
Depth measurement accuracy	[0.09+(0.013depth)^2]1/2m,2σ
Horizontal measurement accuracy	(3.5+0.05depth)m, 2σ
Sounding density	2 mx2 m nominal
Sean angle	20°(fixed off-nadir, circular pattern)
Swath width	294 m nominal

i) The reference bathymetry derived from the GEBCO_2025 Grid is illustrated in Fig. 3, covering (a) Oahu, (d) coastal Xuwen Reef, (e) Dongpingzhou Island, and (f) Saipan, obtained from www.gebco.net. The actual water depth data for the study area (c) Weizhou Island in Fig. 6 was acquired on November 19, 2019, using the RESON SeaBat 7125 multibeam echo sounder system. Its specific measurement accuracy and error limits are detailed in Tables II and III. The actual bathymetric data for the study area (b), Molokai Island, in Fig. 6, were acquired on December 15, 2003, using the SHOAL-3000 laser scanning system. Its key parameters are listed in Table IV. Water depth data from all six study areas were converted to a 30-m spatial resolution consistent with Landsat-8 imagery using ArcGIS software.

ii) *Tidal correction data:* The remote sensing imagery in this study spans multiple tidal cycles and includes typical tidal states from low tide to high tide. The corresponding tidal information primarily originates from tide gauge stations surrounding the study area, including Weizhou Island and Liusha stations along China's coast [45], Honolulu and Kona Stations in the Hawaiian Islands of the United States, and Saipan Port station in the western Pacific [46], as well as

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TABLE V
DATASETS USED FOR VALIDATION

Dataset Type	Images Number	Source Regions	Primary Purpose
Training Set	60	15 images randomly selected from each of 4 regions: Molokai Island, Weizhou Island, Saipan Island, and the coastal area of Xuwen	For model training
Internal Validation Set	20	Remaining images from the aforementioned 4 regions (5 images per region)	Monitor the model's performance on unseen data with the same geographical distribution
External Validation Set	40	Two completely geographically independent new regions: Oahu Island and Dongpingzhou Island	Evaluate the model's generalization ability and robustness in new environments

TABLE VI
EXPERIMENTAL ENVIRONMENT SETTINGS AND TRAINING PARAMETERS

Parameters	Configuration
Operating system	64-bit x86 Linux System
CPU	Intel® Xeon® Silver 4216 (CPU@2.10GHz)
GPU	RTX 4090D
GPU memory size	24G
Implementation Language	Python
Core Libraries	PyTorch 2.5.1+cu124, mamba 2.2.5
Hyper parameters	**Value**
Training Epochs	100
Batch Size	4
Optimizer	Adam
Learning Rate	0.001
Total Number of Model Parameters	27.062million

Gaolauwan Station in Hong Kong [47]. Finally, this study employs the TPXO9 global tidal model to derive hourly tidal predictions during satellite transits [15], [48]. All tidal data are uniformly converted to the vertical reference surface of mean sea level, thereby eliminating systematic differences arising from differing reference frames.

C. Experiment and Analysis

1) *Experimental Configuration:* To ensure a fair evaluation, the dataset is divided into training, internal validation, and external validation subsets, as summarized in Table V. All compared models are trained and evaluated under the same experimental settings, and the implementation environment is reported in Table VI. Due to differences among the six study areas, all input data are normalized before training to unify the data scale and facilitate model convergence. After inference, inverse normalization is applied to recover physically meaningful depth values for quantitative evaluation.

2) *Weight Optimization:* A correlation-based spectral weighting mechanism is applied to the seven subsurface spectral bands to enhance depth-sensitive signals before feature extraction. Each band weight w_i is derived from the averaged Spearman and Wilson correlation coefficients and then normalized. The normalized weight is calculated as follows:

$$w_i = \frac{\bar{R}_i}{\sum_{j=1}^{7} \bar{R}_i}. \tag{11}$$

TABLE VII
OPTIMIZED INPUT WEIGHTS DERIVED FROM SEVEN SPECTRAL FEATURES

Band Name	Median Combined Correlation ($\rho_{combined}$)	Normalized Weight ($Weight_i$)
B1: Coastal Aerosol	0.63	0.1665
B2: Blue	0.62	0.1672
B3: Green	0.62	0.1673
B4: Red	0.59	0.1593
B5: Near Infrared	0.50	0.1104
B6: SWIR 1	0.49	0.1128
B7: SWIR 2	0.49	0.1165

The term $\bar{R}$ represents the mean value of the two correlation types [49]

$$\bar{R} = \frac{R_{Spearman} + R_{Wilson}}{2}. \tag{12}$$

As shown in Table VII and Fig. 7, visible bands (B1–B4) exhibit higher median correlations above 0.59. These bands receive large normalized weights due to their superior water penetration. Conversely, the weighting mechanism suppresses the noisy NIR/SWIR bands (B5–B7). These specific bands show lower median correlations and greater dispersion in the violin plots.

This strategy effectively reduces background interference for the deep network. It prioritizes robust subsurface spectral information during the initial training phase.

3) Result of Bathymetry Retrial With the Proposed Method: To further examine the ability of SDB-Mamba to reconstruct continuous depth variations, check-line profiles were extracted from the six study areas and compared with the corresponding ground-truth (GT) bathymetry, as shown in Figs. 8–13. These profiles provide an intuitive assessment of both vertical accuracy and morphological consistency along representative depth gradients.

Overall, the predicted profiles agree well with the GT curves across different coastal environments. In relatively stable optical conditions, such as Molokai Island and Xuwen County, SDB-Mamba closely follows the observed depth variations, achieving high agreement with the reference profiles.

The results show strong statistical consistency, with representative values of R^2 = 0.99 and MAE = 0.64 m for Molokai Island, and R^2 = 0.98 and MAE = 0.18 m for Xuwen County. These results indicate that the proposed model can accurately recover smooth depth transitions and low-amplitude seabed variations.

For more complex regions, including Oahu Island, Saipan Island, and Tung Ping Chau Island, the model still captures the major bathymetric patterns, such as multistage depth changes, platform–trough structures, and steep slope transitions. Local discrepancies are mainly observed near abrupt geomorphic boundaries or sharp inflection points, for example, around Pixel 150 in Molokai, Pixel 157 in Oahu, Pixel 153 in Saipan, and Pixel 71 in Tung Ping Chau.

These deviations are likely related to rapid terrain changes, optical attenuation, and heterogeneous bottom reflectance, which can weaken the correspondence between spectral response and water depth.

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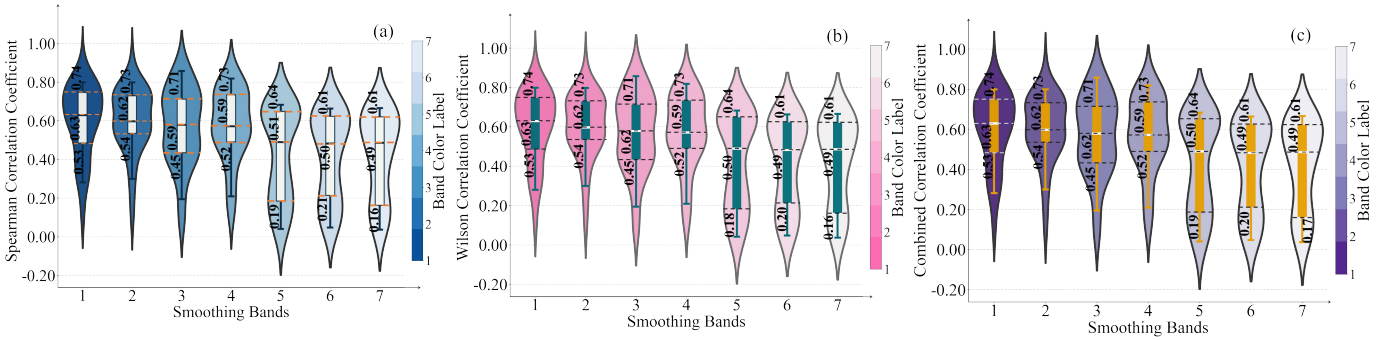


Fig. 7. Distribution violin plot for (a) Sparman correlation, (b) Wilson correlation, and (c) combined correlation.

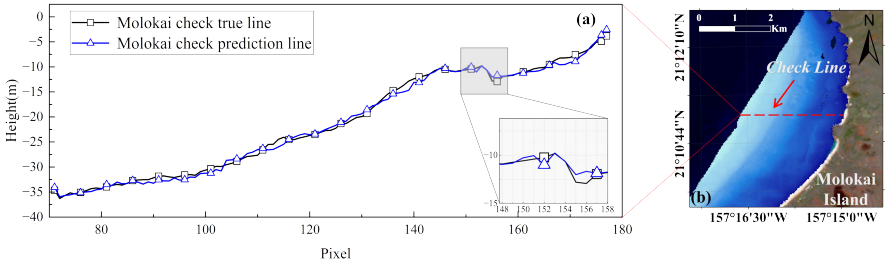


Fig. 8. MolokaiIsland bathymetric check line. (a) True and predicted bathymetric profiles. (b) Location of the selected check line.

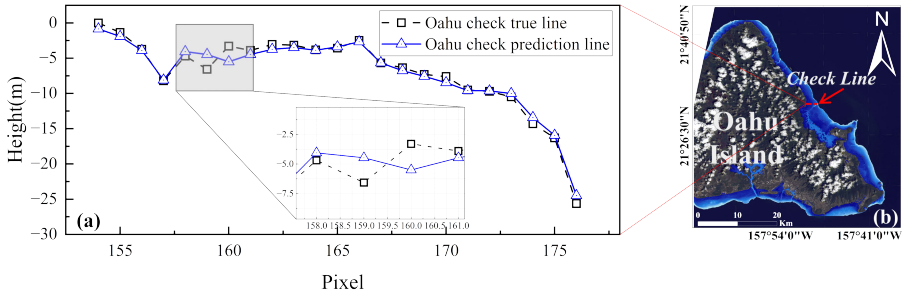


Fig. 9. Oahu Island bathymetric check line. (a) True and predicted bathymetric profiles. (b) Location of the selected check line.

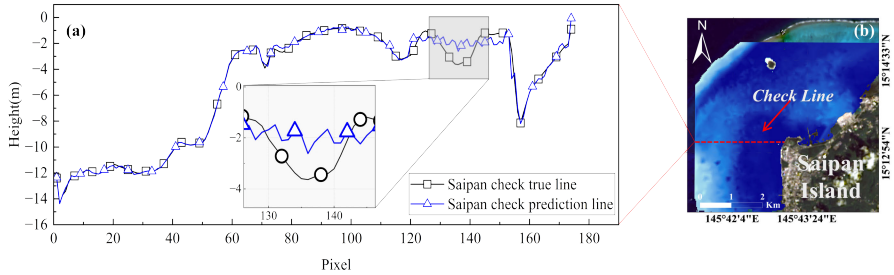


Fig. 10. Saipan Island bathymetric check line. (a) True and predicted bathymetric profiles. (b) Location of the selected check line.

In Weizhou Island, the predicted profile follows the monotonic offshore-to-nearshore shoaling trend with good consistency, indicating stable performance in areas with smoother morphology and relatively uniform optical conditions. Across all six profiles, the model preserves the overall shape of the GT curves while maintaining limited local bias. This confirms that SDB-Mamba can reconstruct both large-scale bathymetric gradients and local terrain variations, although sharp transitions and optically heterogeneous zones remain challenging.

Overall, the profile-based evaluation demonstrates that SDB-Mamba provides reliable and spatially coherent bathymetric reconstruction along representative transects. The remaining errors are mainly concentrated near steep geomorphic transitions or regions with complex spectral responses, which are consistent with the known limitations of SDB under depth-dependent optical attenuation.

4) Comparison With SOTA Models: To comprehensively evaluate the performance of the proposed framework, comparative experiments were conducted against eight representative

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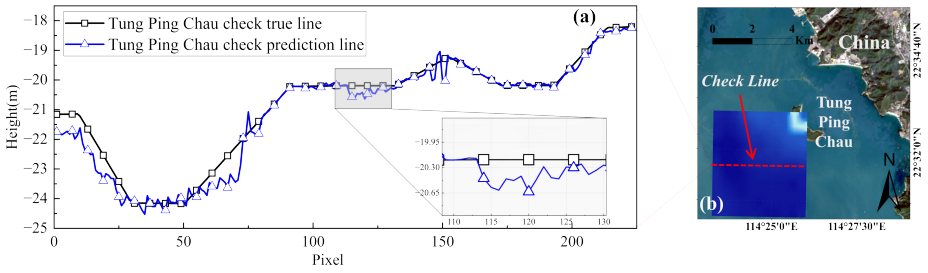


Fig. 11. Tung Ping Chau Island bathymetric check line. (a) True and predicted bathymetric profiles. (b) Location of the selected check line.

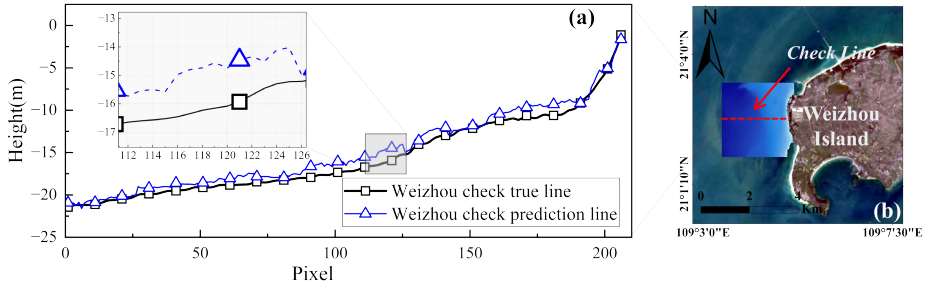


Fig. 12. Weizhou Island bathymetric check line. (a) True and predicted bathymetric profiles. (b) Location of the selected check line.

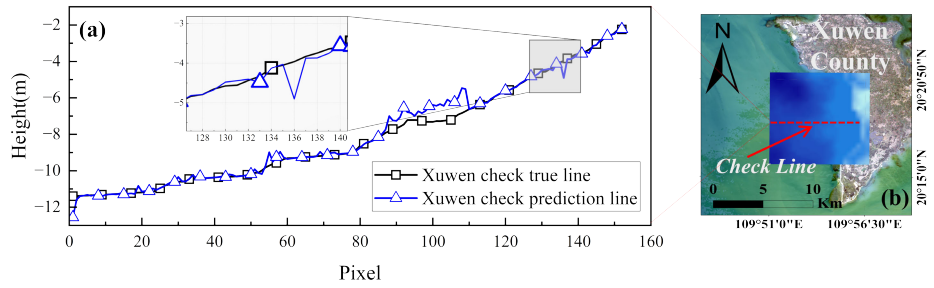


Fig. 13. Xuwen County bathymetric check line. (a) True and predicted bathymetric profiles. (b) Location of the selected check line.

TABLE VIII
DL EVALUATION ACROSS SIX STUDY AREAS

Method	MAE (m) ↓ (Mean±std)						RMSE (m) ↓ (Mean±std)					
	Area 1 internal test	Area 2 internal test	Area 3 internal test	Area 4 internal test	Area 5 external test	Area 6 external test	Area 1	Area 2	Area 3	Area 4	Area 5	Area 6
U-Net	1.213±0.348	1.585±0.230	1.795±0.356	2.516±0.565	1.466±0.040	4.643±0.649	2.388±1.302	2.341±0.358	2.232±0.450	3.389±0.684	3.518±0.133	5.384±0.721
SwinUnet	2.890±0.696	2.364±0.379	3.631±0.575	3.164±0.458	1.830±0.073	5.413±0.950	5.425±1.294	3.963±0.607	4.588±0.743	4.179±0.518	4.394±0.100	7.690±1.084
VMUNet	0.612±0.246	0.605±0.014	0.914±0.560	0.686±0.080	0.791±0.041	0.616±0.053	1.440±1.194	0.930±0.031	1.141±0.726	0.956±0.087	1.182±0.185	0.881±0.156
H-VMUNet	0.814±0.242	0.736±0.052	1.383±1.380	0.569±0.262	1.200±0.082	0.762±0.364	1.801±0.712	1.266±0.032	1.528±1.493	0.720±0.287	2.850±0.173	1.026±0.515
TransUNet	2.565±0.865	2.415±0.543	2.472±0.363	3.175±0.560	1.705±0.134	4.717±1.661	4.189±1.686	3.454±0.701	3.100±0.467	3.845±0.581	3.578±0.035	5.757±1.841
SwinBathyUNet	1.416±0.398	2.041±0.441	1.949±0.503	2.813±0.766	1.788±0.046	3.174±0.826	2.930±1.420	3.312±0.736	2.466±0.598	3.642±0.981	3.842±0.124	3.958±0.688
MSFMamba	0.788±0.160	1.207±0.318	1.028±0.470	0.822±0.092	1.136±0.021	1.198±0.521	1.417±0.169	1.826±0.359	1.223±0.439	1.073±0.097	3.044±0.101	1.793±1.018
Mamba Fusion	1.507±0.559	1.533±0.239	2.077±0.338	2.375±0.711	1.319±0.067	2.820±0.905	2.165±0.667	2.260±0.335	2.618±0.340	3.136±0.759	3.116±0.153	3.642±0.992
Proposed model	**0.537±0.089**	**0.524±0.022**	**0.562±0.327**	**0.638±0.264**	**0.360±0.044**	**0.531±0.068**	**1.406±0.609**	**0.885±0.012**	**0.698±0.390**	**0.834±0.317**	**0.937±0.096**	**0.721±0.170**
Method	SSIM ↑ (Mean±std)						PSNR (dB) ↑ (Mean±std)					
	Area 1	Area 2	Area 3	Area 4	Area 5	Area 6	Area 1	Area 2	Area 3	Area 4	Area 5	Area 6
U-Net	0.670±0.057	0.630±0.037	0.759±0.031	0.488±0.079	0.634±0.036	0.832±0.033	27.534±3.899	24.785±1.309	19.895±1.586	14.889±1.886	23.244±0.323	14.882±1.311
SwinUnet	0.641±0.092	0.571±0.041	0.633±0.038	0.367±0.073	0.476±0.030	0.670±0.045	19.678±2.192	20.208±1.228	13.591±1.601	12.954±1.222	21.309±0.196	11.784±1.076
VMUNet	**0.905±0.028**	0.830±0.012	0.846±0.063	0.688±0.079	0.882±0.008	0.924±0.008	**32.971±5.308**	**33.141±0.183**	26.865±4.294	27.363±2.889	30.415±0.699	30.652±1.521
H-VMUNet	0.822±0.057	0.801±0.016	0.881±0.080	0.671±0.081	0.768±0.018	0.910±0.019	29.518±2.858	30.026±0.261	25.957±6.345	28.702±3.046	25.083±0.510	30.234±4.337
TransUNet	0.467±0.077	0.531±0.065	0.717±0.042	0.422±0.052	0.524±0.206	0.772±0.049	22.372±3.537	21.483±1.742	16.973±1.299	13.712±1.455	23.090±0.084	14.593±2.414
SwinBathyUNet	0.744±0.028	0.607±0.026	0.751±0.025	0.502±0.060	0.490±0.010	0.841±0.020	25.599±3.635	21.861±1.751	19.119±2.092	14.374±2.272	22.476±0.276	17.593±1.383
MSFMamba	0.745±0.079	0.665±0.053	0.883±0.012	0.579±0.069	0.820±0.006	0.893±0.033	31.255±1.066	27.053±1.640	25.477±2.677	24.578±0.674	24.493±0.287	25.412±4.058
Mamba Fusion	0.544±0.096	0.613±0.052	0.788±0.029	0.406±0.050	0.803±0.014	0.768±0.042	27.724±2.340	25.135±1.298	18.415±1.354	15.456±2.093	24.295±0.413	18.501±2.240
Proposed model	0.884±0.016	**0.844±0.038**	**0.904±0.012**	**0.763±0.056**	**0.965±0.006**	**0.943±0.006**	31.758±3.054	32.014±0.197	**27.862±3.850**	**28.738±0.687**	**35.074±0.310**	**32.467±1.074**

state-of-the-art models, including U-Net [50], [51], SwinUNet [52], VMUNet [38], H-VMUNet [53], TransUNet [54], SwinBathyUNet [33], MSF-Mamba [55], and Mamba Fusion [56]. The evaluation was performed across six study areas, including Molokai Island, Saipan Island, Weizhou Island, Xuwen County, Oahu Island, and Tung Ping Chau Island. Several commonly used metrics were adopted, including R^2, RMSE [57], MAE [58], structural similarity index measure (SSIM) [59], and peak signal-to-noise ratio (PSNR) [60]. The quantitative results are summarized in Table VIII.

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TABLE IX
COMPARATIVE ANALYSIS OF MODEL PARAMETERS, FLOPS AND INFERENCE TIME

Metrics	U-Net	SwinUnet	VMUNet	H-VMUNet	TransUNet	SwinBathy UNet	MSFMamba	Mamba-SDB Fusion	Proposed model
Params (M)	31.046	27.152	22.447	6.439	66.840	51.899	6.598	6.098	27.062
FLOPS (G)	42.024	5.930	4.0314	0.582	25.309	64.895	9.770	4.039	4.182
Inference times (s)	4.563	14.292	47.994	102.545	8.537	16.013	19.665	6.563	18.451

Overall, SDB-Mamba achieves highly competitive performance across different regions and metrics. In terms of vertical accuracy, the proposed model shows clear advantages in most study areas. Compared with VMUNet, which is the second-best model in several cases, SDB-Mamba reduces MAE by 12.3% in Area 1% and 13.4% in Area 2. In the external test area of Oahu Island, the MAE reduction reaches 54.5%, indicating stronger cross-scene generalization.

It should be noted that SDB-Mamba does not achieve the best value for every individual metric. For example, VMUNet obtains a slightly higher SSIM in Area 1, while H-VMUNet shows competitive structural consistency in some complex regions. This phenomenon is reasonable because SSIM and PSNR are sensitive to local texture similarity, and simpler or more locally biased models may preserve certain spatial structures more effectively in relatively homogeneous areas. Nevertheless, SDB-Mamba provides more consistent overall performance, especially in RMSE and MAE, which are directly related to vertical bathymetric accuracy. These results indicate that the proposed method achieves a better balance between depth precision, structural fidelity, and cross-region robustness.

5) *Computational Efficiency Analysis:* To evaluate the engineering feasibility of SDB-Mamba for large-scale bathymetric mapping, model parameters, computational cost, and inference time were compared, as reported in Table IX. The results show that SDB-Mamba achieves a favorable balance between accuracy and efficiency.

SDB-Mamba contains 27.062M parameters. Although this value is higher than those of some lightweight models, it is much lower than that of Transformer-based models. For example, the parameter count is reduced by 59.5% compared with TransUNet, which has 66.840M parameters. In terms of computational complexity, SDB-Mamba requires only 4.182G FLOPs, corresponding to reductions of 93.6% and 83.5% compared with SwinBathyUNet and TransUNet, respectively.

The total inference time of SDB-Mamba on the test dataset is 18.451 s. This value is slightly higher than that of Mamba Fusion, but clearly lower than those of VMUNet and H-VMUNet, which requires 47.994 and 102.545 s, respectively. Compared with H-VMUNet, SDB-Mamba reduces inference time by 82.0%. Combined with the accuracy results in Table VIII, these results suggest that the performance gain of SDB-Mamba does not rely on excessive model scale, but instead benefits from an efficient architectural design based on selective scanning and cross-scale feature interaction.

6) *Stratified Error Analysis Across Depth Intervals:* To further analyze model stability under depth-dependent optical conditions, a stratified error analysis was conducted across four depth intervals: –0 to –10 m, –10 to –20 m, –20 to –30 m, and –30 to –40 m. The corresponding results are shown in Figs. 14 and 15.

All models show an increasing trend in MAE and RMSE as water depth increases. This trend is mainly caused by the attenuation of bottom-reflected signals and the reduction in signal-to-noise ratio. In the shallow-water interval of –0 to –10 m, most models achieve relatively low and comparable errors because the spectral response remains strong. However, the differences among models become more evident in deeper intervals.

In the –20 to –40 m ranges, several baseline models, including CNN-based and Transformer-based architectures, show more noticeable performance degradation. In contrast, SDB-Mamba maintains a more stable error increase across depth intervals. In the deepest interval of –30 to –40 m, the proposed method still achieves lower MAE than the baseline models and shows a relative improvement over VMUNet. These results indicate that correlation-guided spectral modulation and selective scanning help preserve useful spectral information under low-signal conditions. Although spectral saturation remains a physical limitation, SDB-Mamba provides more consistent performance across different depth ranges.

7) *Qualitative Evaluation:* Spatial Mapping and Scatter Analysis. To further evaluate spatial consistency and statistical reliability, bathymetric inversion maps and density scatter plots for the six study areas are presented in Figs. 16 and 17, respectively. These results provide complementary evidence from spatial morphology and pointwise regression consistency.

As shown in Fig. 16, SDB-Mamba produces spatially coherent bathymetric maps across different coastal environments. Compared with baseline models, the proposed method better preserves continuous depth transitions and local geomorphological structures, such as reef edges, shallow-water platforms, and slope variations. Some baseline methods exhibit patchy artifacts, block-like patterns, or over-smoothed gradients, especially in optically complex or deeper regions. In contrast, SDB-Mamba maintains smoother large-scale depth patterns while retaining important local details.

The density scatter plots in Fig. 17 further supports this observation. Across the six study areas, the predicted depths of SDB-Mamba are generally concentrated around the 1:1 reference line, indicating good agreement with the reference bathymetry. Compared with competing methods, the proposed model shows a tighter point distribution and fewer obvious outliers in most regions. This trend is consistent with the quantitative results in Table VIII, where SDB-Mamba achieves lower RMSE and MAE in most cases. Overall, the spatial maps and scatter distributions indicate that SDB-Mamba

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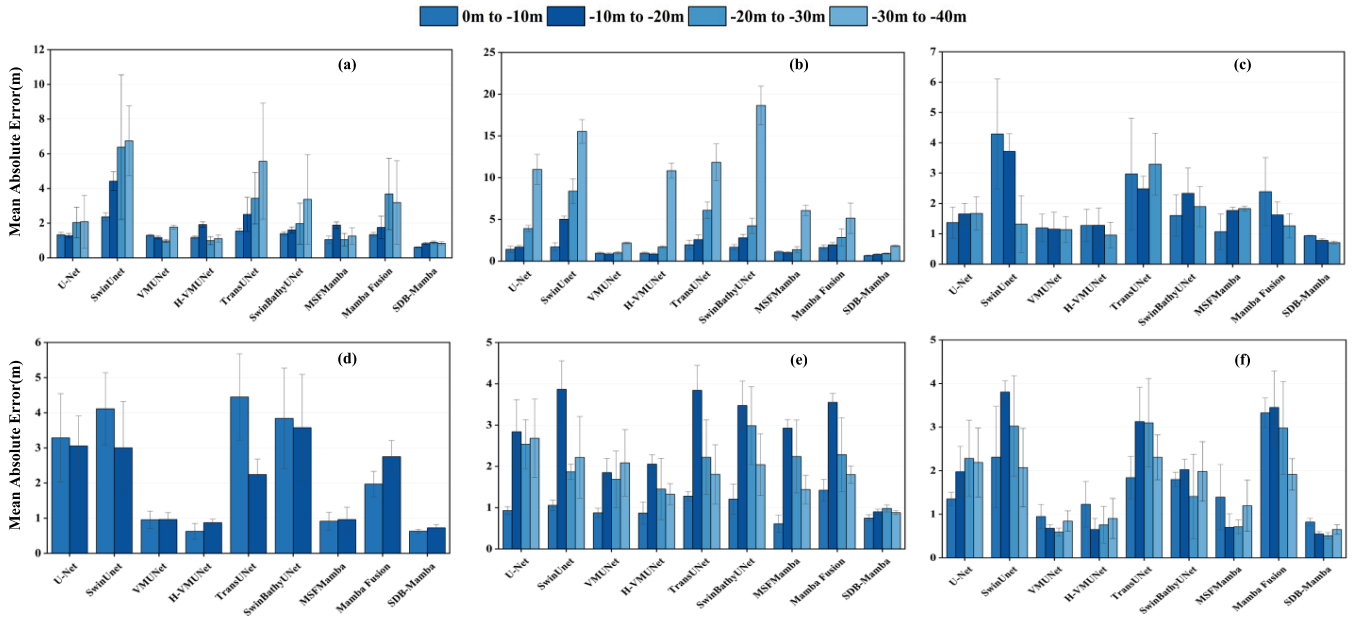


Fig. 14. Depth-stratified MAE (↓) comparison across six study areas. (a) Study area 1. (b) Study area 2. (c) Study area 3. (d) Study area 4. (e) Study area 5. (f) Study area 6.

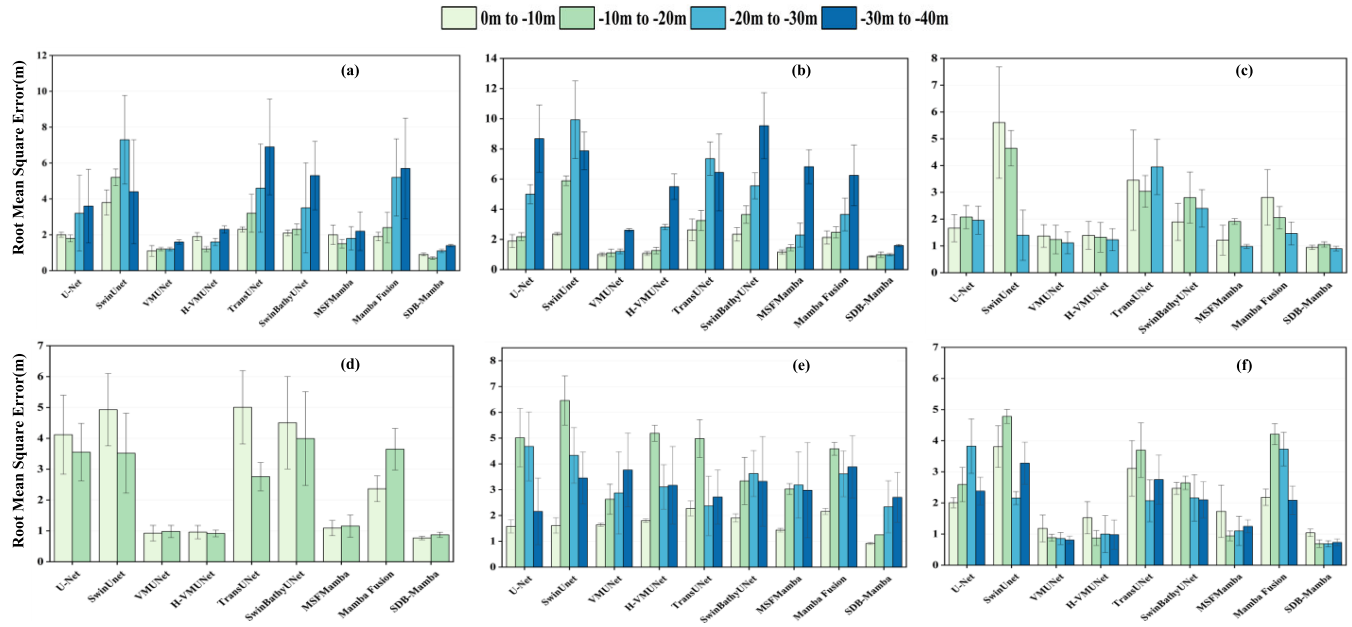


Fig. 15. Depth-stratified RMSE (↓) comparison across six study areas. (a) Study area 1. (b) Study area 2. (c) Study area 3. (d) Study area 4. (e) Study area 5. (f) Study area 6.

improves both spatial continuity and regression consistency, while the remaining errors mainly occur near sharp geomorphic transitions or regions with heterogeneous optical conditions.

D. Ablation Study

1) Ablation Study Setup: To evaluate the contribution of each component, a controlled ablation study was conducted, as reported in Table X. The baseline model uses only the VMamba encoder without skip connections or attention modules. Skip connections, channel attention, spatial attention, cross attention, and different backbone combinations were then progressively introduced. The ablation also examines the execution order of attention modules, where the "✓first" mark indicates the module applied first. MAE, RMSE, PSNR, and SSIM are reported as mean ± standard deviation.

2) Analysis of Ablation Study Results: The results in Table X show that each proposed component contributes to the final performance. The single VMamba baseline obtains the weakest performance, with an MAE of 7.321. After introducing skip connections, the MAE decreases to 2.099,

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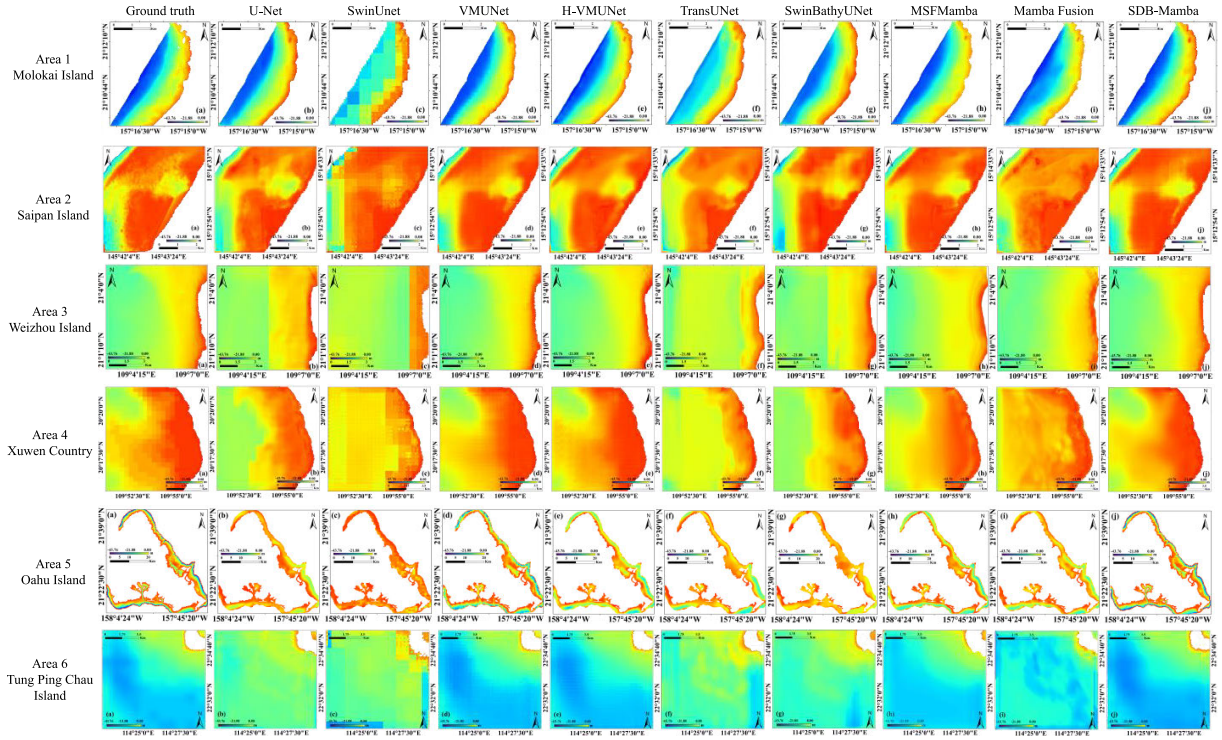


Fig. 16. Comparison of bathymetric inversion maps across six study areas. (a) Ground Truth. (b) Predictions of U-Net. (c) Predictions of SwinUnet. (d) Predictions of VMUNet. (e) Predictions of HVMUNet. (f) Predictions of TransUNet. (g) Predictions of SwinBathyUNet. (h) Predictions of MSFM amba. (i) Predictions of Mamba Fusion. (j) Predictions of SDB-Mamba (ours).

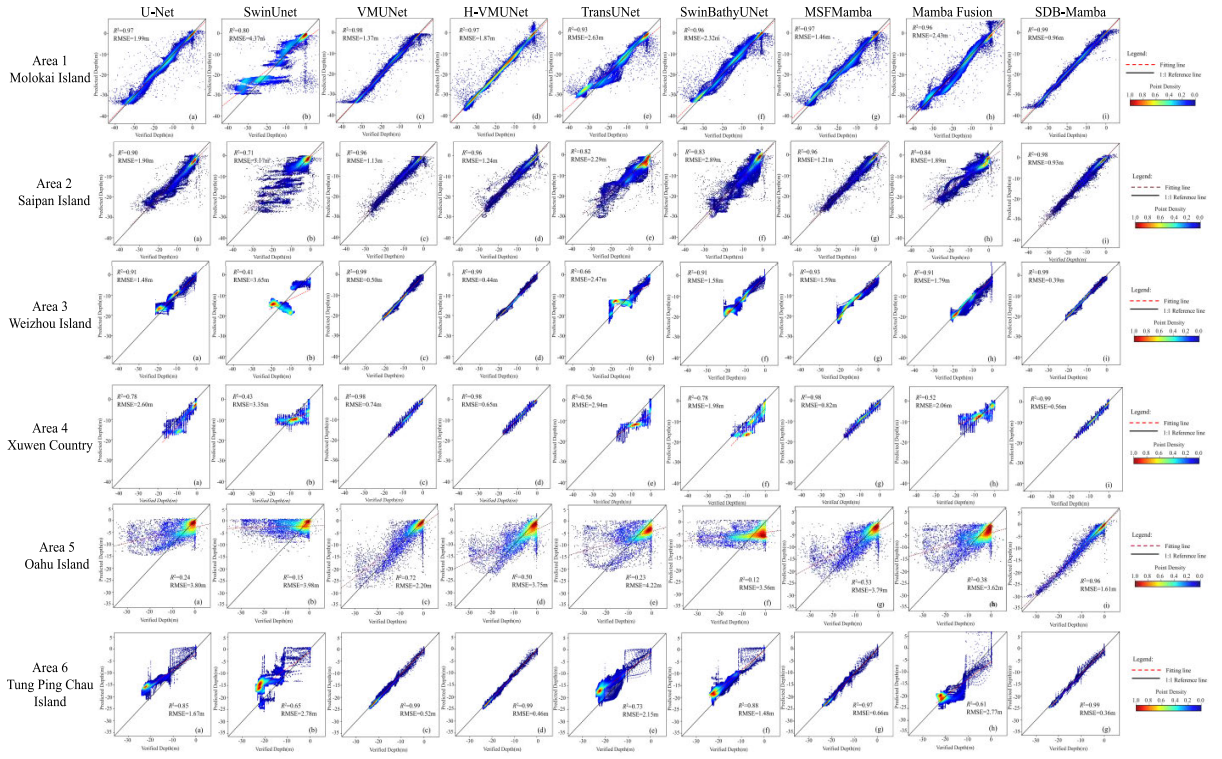


Fig. 17. Density scatter plots of predicted versus reference bathymetry across six study areas. (a) Ground Truth. (b) Predictions of U-Net. (c) Predictions of SwinUnet. (d) Predictions of VMUNet. (e) Predictions of HVMUNet. (f) Predictions of TransUNet. (g) Predictions of SwinBathyUNet. (h) Predictions of MSFM amba. (i) Predictions of Mamba Fusion. (j) Predictions of SDB-Mamba (ours).

corresponding to a 71.3% reduction. This result indicates that cross-layer feature compensation can effectively reduce information loss during deep feature propagation.

The attention modules further improve the reconstruction accuracy. Using channel attention or spatial attention alone brings performance gains, while their combination provides

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TABLE X
ABLATION 11 EXPERIMENTS PERFORMANCE

ResNet Encoder	VMamba Encoder	Skip Connect	Add Connect	Cross Attention	Channel Attention	Spatial Attention	MAE↓ (mean±std)	RMSE↓ (mean±std)	PSNR↑ (mean±std)	SSIM↑ (mean±std)
×	✓	×					7.321 ± 1.783	8.473 ± 2.958	11.857 ± 3.013	0.482 ± 0.199
×	✓	✓	✓				2.099 ± 0.743	2.013 ± 1.396	26.691 ± 4.899	0.808 ± 0.208
×	✓	✓		✓			2.229 ± 0.830	3.783 ± 1.305	22.537 ± 4.697	0.756 ± 0.071
×	✓	✓			✓		1.901 ± 1.487	2.783 ± 2.306	20.508 ± 4.956	0.779 ± 0.187
×	✓	✓				✓	1.824 ± 1.663	1.886 ± 2.571	21.130 ± 2.969	0.812 ± 0.204
×	✓	✓			✓first	✓	1.739 ± 1.467	1.963 ± 2.142	26.445 ± 4.428	0.822 ± 0.185
×	✓	✓			✓	✓first	1.603 ± 0.693	1.724 ± 1.499	27.120 ± 4.044	0.844 ± 0.068
✓	×	✓					1.048 ± 0.749	1.550 ± 1.198	28.872 ± 4.576	0.831 ± 0.056
✓ (NW)	✓ (NW)	✓			✓	✓first	2.056 ± 1.069	1.923 ± 1.102	25.695 ± 4.801	0.782 ±0.188
✓(NSSRM)	✓	✓			✓	✓first	1.957 ± 1.857	2.128 ± 2.051	22.778 ± 4.352	0.684± 0.196
✓	✓	✓			✓	✓first	**0.954 ± 0.500**	**1.299 ± 0.712**	**28.932 ± 4.808**	**0.849 ± 0.092**

better results. In particular, the Spatial → Channel order achieves a strong intermediate performance, with an MAE of 1.603 and an SSIM of 0.844. Compared with the baseline, this setting reduces MAE by 78.1% and improves SSIM by 75.1%. This suggests that spatial localization and channel recalibration are complementary for bathymetric feature refinement.

The complete dual-encoder architecture achieves the best overall performance, with an MAE of 0.954, RMSE of 1.299, and PSNR of 28.932. Compared with single-backbone variants, the combination of ResNet and VMamba provides better balance between local spatial detail extraction and long-range dependency modeling. Relative to the baseline, the full SDB-Mamba reduces MAE by 86.9%, demonstrating the effectiveness of integrating residual convolution, Vision Mamba, and dual attention within a unified framework.

Overall, the ablation results confirm the complementarity of the proposed modules and the effectiveness of the final architecture.

IV. DISCUSSION

A. Influence of Each Module

1) Vision Mamba: The Vision Mamba module enables efficient long-range dependency modeling through state-space representations, avoiding the quadratic complexity of self-attention. Compared with CNN-based encoders that mainly capture local patterns, and transformer-based models that incur high computational cost, Vision Mamba provides a more efficient balance between global context modeling and scalability. As reflected in the quantitative results (Table VIII) and ablation study (Table X), the inclusion of Vision Mamba consistently reduces RMSE and MAE across different study areas. This improvement suggests that modeling long-range spectral–spatial dependencies help mitigate prediction uncertainty caused by variations in water optical properties and heterogeneous reflectance conditions.

2) ResNet34: ResNet34 complements the Vision Mamba branch by providing robust local feature extraction. It effectively captures fine-grained spatial structures, such as shoreline boundaries, shallow-water textures, and sediment patterns, which are closely related to depth variations. The residual connections also ensure stable feature propagation during training. As shown in the ablation results, removing the ResNet branch leads to noticeable performance degradation, indicating that local spatial cues remain essential. The combination of ResNet34 and Vision Mamba therefore enables the model to jointly exploit local detail and global context, improving both convergence stability and spatial accuracy.

3) Dual Attention Mechanism: The dual attention module further refines feature representations by combining spatial and channelwise recalibration. The spatial attention branch highlights regions with significant depth-related variation, while the channel attention branch adaptively emphasizes informative spectral bands. As demonstrated in the ablation study, the joint use of spatial and channel attention consistently outperforms single-attention configurations. This result indicates that the coordinated modeling of spectral and spatial importance is critical for handling complex underwater environments, especially in regions affected by turbidity gradients, bottom heterogeneity, and optical noise [61].

B. Uncertainty of Reference Bathymetry

The interpretation of retrieval accuracy is inherently affected by the uncertainty of the reference bathymetric data. In this study, the ground truth is derived from multiple sources, including in situ measurements, LiDAR surveys, and chart-based products, each with different levels of accuracy, spatial resolution, and georeferencing consistency. These uncertainties may propagate into the evaluation metrics and impose a practical lower bound on the achievable error.

In optically clear shallow-water regions, where the signal-to-noise ratio is relatively high, the model may produce very low retrieval errors that approach, or in some cases appear lower than, the effective uncertainty of the reference data. This phenomenon does not imply that the model exceeds the intrinsic accuracy of the ground truth; rather, it reflects the fact that both the predictions and the reference data contain uncertainty.

Therefore, the reported accuracy should be interpreted with caution, particularly in regions where reference uncertainty is nonnegligible. Future work should incorporate uncertainty-aware evaluation strategies and higher-precision reference datasets to better assess the reliability of SDB.

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C. Limitations and Future Works

Despite its advantages, the proposed framework has several limitations. First, although the Mamba module effectively models long-range dependencies, its internal state transition mechanism lacks interpretability, particularly in relation to underwater radiative transfer processes. Second, the dual-encoder architecture increases model complexity, which may limit its applicability in resource-constrained scenarios or small-sample settings. In addition, attention mechanisms may amplify noise in optically complex regions, such as highly turbid waters or areas affected by sun glint, leading to local prediction instability.

Future work will focus on improving both robustness and interpretability. One direction is to incorporate multisource or multitemporal remote sensing data, such as Sentinel-2 and UAV imagery, to enhance cross-environment generalization. Another direction is to integrate physical constraints or radiative transfer priors to better link data-driven representations with underlying optical processes. In addition, lightweight model design, including pruning and knowledge distillation, will be explored to support real-time and edge deployment. From a theoretical perspective, further investigation into the interaction between state-space modeling and spectral reflectance physics may provide a more generalizable framework for bathymetric inversion.

V. Conclusion

This study proposes SDB-Mamba, a dual-encoder–single-decoder framework for SDB using subsurface reflectance. The method introduces correlation-aware spectral modulation to emphasize depth-sensitive bands and uses a ResNet34 branch with a SSRM to preserve local spectral–spatial details. In parallel, the Vision Mamba branch captures long-range dependencies in subsurface reflectance, enabling more effective modeling of large-scale bathymetric variations.

A dual-attention fusion bridge and a Mamba-based decoder are further designed to enhance feature interaction during reconstruction. The spatial–channel attention mechanism improves sensitivity to bathymetry-related structures while suppressing redundant responses, thereby improving the stability and spatial coherence of depth prediction.

Experiments across multiple study areas show that SDB-Mamba achieves competitive and consistent performance compared with traditional machine learning methods, U-Net-based networks, and transformer-based models. The quantitative comparisons, stratified depth analysis, profile evaluation, and visual results jointly demonstrate its advantages in vertical accuracy, spatial continuity, and cross-region generalization. Ablation studies further confirm the complementary roles of correlation-guided spectral modulation, the ResNet34 encoder, the Vision Mamba encoder, and the dual-attention fusion strategy.

Despite these improvements, the accuracy of SDB remains affected by water optical complexity, depth-dependent signal attenuation, and the uncertainty of reference bathymetric data. Future work will focus on multisource data fusion, uncertainty-aware evaluation, physical-prior integration, and lightweight model design to improve interpretability, robustness, and operational applicability in large-scale coastal and marine mapping.

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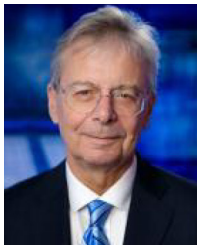
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